

Muhammad Talha Zaheer

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PROFESSIONAL SUMMARY

Computer scientist specializing in Agentic AI and secure software architecture. Experienced in building autonomous AI solutions with Next.js, Node.js, and the Vercel AI SDK—including tool calling, streaming generation, and production deployment. Combines full-stack development with a defensive security mindset shaped by hands-on cybersecurity training and instruction. Passionate about technical education, community leadership, and designing intelligent systems that are both innovative and secure.

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, Bash, SQL
Frameworks	Next.js, React, Node.js
AI & ML	Agentic AI, LLMs, Vercel AI SDK, Tool Calling, Pipecat AI, Deepgram, Cartesia, Daily PSTN, Prompt Optimization, LM Studio
DevOps & Deployment	Git/GitHub, GitHub Actions, Docker, CI/CD, Vercel, Railway, Digital Ocean, Supabase
Tools & IDEs	Postman, Cursor, VS Code, Google Antigravity, VirtualBox
Project Management	Monday.com, Linear

PROFESSIONAL EXPERIENCE

Core Edge Solutions	<i>May 2025 – May 2026</i>
<i>AI Developer</i>	<i>Lahore, Pakistan</i>

- Built production voice agents using Retell and Pipecat AI, integrating Deepgram (STT), OpenAI (LLM), and Cartesia (TTS) for cost-efficient conversational calling.
- Implemented tool calling for call transfer, call end, voicemail detection, and external API integration; optimized Pacifica Bot prompts after learning TMS domain workflows.
- Developed *Zuum* (FastAPI + React), an internal platform that outbound-calls dispatch drivers for shipment status updates.
- Designed GitHub Actions CI/CD pipelines to automate deployments on Railway.
- Built Puppeteer automations for Trinium TMS: adding container notes, updating empty container status, and processing acknowledged containers.
- Wrote SQL queries against Trinium TMS, exposed data through custom APIs, and compiled reports for container tracking.

Professional Freelancing Training Program	<i>Nov 2024 – Apr 2025</i>
<i>Cybersecurity Intern</i>	<i>Lahore, Pakistan</i>

- Conducted web application penetration testing for SQL injection, file inclusion, and unrestricted file upload vulnerabilities.
- Performed network reconnaissance with Nmap and analyzed HTTP/traffic flows using Burp Suite and Wireshark.
- Researched and lab-tested exploits including EternalBlue (MS17-010), macro-based attacks, and GHDB-driven reconnaissance; tracked emerging CVEs for early evaluation.
- Configured WAF rules and pfSense firewalls to protect web applications; practiced digital forensics on hard drives and disk images.

PROJECTS

Vello (Voice Agent)	<i>AI Agent</i>
<i>Tech Stack:</i> Pipecat AI, Deepgram STT, OpenAI LLM, Cartesia TTS, Daily PSTN, Supabase, React	

- Built a Pipecat AI voice bot for inbound and outbound calls via Daily PSTN, with voicemail detection, call transfer, and professional sales-agent conversations.
- Integrated Deepgram STT, OpenAI LLM, and Cartesia TTS; supports separate dynamic system prompts for inbound and outbound agents loaded from Supabase at call time.
- Tracks token consumption for per-call pricing; stores recordings and transcriptions in Supabase with a React dashboard for org setup, member invites, recordings, and cost analytics.

Pacifica Bot (Customer Support Agent)

Voice Agent

Tech Stack: Pipecat AI, Deepgram STT, OpenAI LLM, Cartesia TTS, Daily PSTN, FastAPI, Supabase, React, GitHub Actions, Railway

- Built an inbound Pipecat AI voice agent via Daily PSTN for logistics customer support, with a comprehensive system prompt covering general guidelines and TMS domain knowledge to resolve queries efficiently, including generic inquiries.
- Implemented agent tools as FastAPI endpoints that query the Trinium database with optimized joins for fast real-time lookups; supports ending calls, transferring calls, and sending emails during live conversations.
- Tracks per-call cost across LLM, TTS, STT, and telephony usage; stores call recordings and transcripts in Supabase for later analysis and review.
- Designed GitHub Actions workflows to deploy on Railway with separate pipelines for production, staging, and per-PR preview environments.

Zuum (Truck Dispatch Agent)

Voice Agent

Tech Stack: Retell AI, Twilio, FastAPI, React, Supabase, WebSockets

- Built a truck dispatch voice agent using Retell AI and Twilio to outbound-call drivers for load status updates, delivery confirmation, hurdle reporting, and POD collection.
- Implemented Retell single-prompt agents with custom functions to fetch dispatch data from a FastAPI backend; used post-call analysis and webhook events to track live and completed calls.
- Stored call data, analysis, and transcriptions in Supabase; React frontend displays dispatch activity with real-time updates via WebSockets.

Automated SDR (Outbound Calling Agent)

Voice Agent

Tech Stack: Pipecat AI, Deepgram STT, OpenAI LLM, Cartesia TTS, Daily PSTN, Supabase, React, GitHub Actions, Railway

- Built an outbound Pipecat AI voice agent for sales development, placing automated calls for prospecting, lead qualification, and follow-up at scale via Daily PSTN.
- Integrated Deepgram STT, OpenAI LLM, and Cartesia TTS with dynamic SDR prompts loaded from Supabase; includes voicemail detection and professional sales conversation flows.
- Tracks token usage for per-call cost analysis; stores recordings and transcriptions in Supabase with a React dashboard for campaign monitoring and call review.
- Designed GitHub Actions workflows to deploy on Railway with separate pipelines for production, staging, and per-PR preview environments.

Agentic Portfolio

AI Web App

Tech Stack: Next.js, Vercel AI SDK, OpenAI, Vercel

- Built a Next.js portfolio app powered by the Vercel AI SDK and OpenAI, using a user-supplied JSON CV as dynamic context for the chatbot.
- UI adapts automatically from the uploaded JSON profile; the agent answers role-fit and project-related queries from that resume data.
- Designed for easy deployment on Vercel with minimal setup—users only need to provide their CV JSON to launch a personalized agentic portfolio.

DrayDog – Puppeteer Automations

Automation Platform

Tech Stack: Node.js, Express, Puppeteer, OpenAI

- Built a Node.js automation platform that fetches container data from DrayDog and runs concurrent Puppeteer workflows for logistics operations.

- Automated marking containers as picked up, updating leg movement status, adding driver notes, and flagging empty containers as available for pickup.
- Collected and stored DrayDog and Trinium data for cross-system comparison; used OpenAI to analyze empty-container emails.

AstroPOS

Restaurant Management System

Tech Stack: Electron, Vite, React, Supabase

- Built a complete restaurant management system with PIN login, multi-kitchen setup, menu items, deals, custom shift tracking, and per-item sales reporting.
- Supports multiple payment accounts, on-account customer billing, auto-saved customer details, delivery charges, and discounts via a full admin panel.
- Works offline and online with interval-based sync; routes orders to multiple printers with separate customer and kitchen receipts; includes an on-screen keyboard and auto-update support.

Chotu Cafe Mobile

Mobile App

Tech Stack: Expo, React Native, Supabase

- Built a waiter-facing mobile app in Expo with Supabase for managing table orders across restaurant service workflows.
- Uses Supabase Realtime to push live order updates as changes are made or new orders are placed.
- Sends remote print instructions via a print server to route receipts to kitchens and customers when needed.

Astro CTF

CTF Platform

Tech Stack: Django, PostgreSQL, Docker, Jinja2

- Built a complete CTF platform in Django with modular apps, PostgreSQL backend, and event lifecycle management (start/end scheduling).
- Supports tasks with encrypted flags, files, links, hints, and Docker-based challenges added via a guided admin UI.
- Tracks team and per-task performance metrics; uses Jinja2 templating to render structured frontend views from large datasets.

File Sync System

C++ Application

Tech Stack: C++, WinSock, AVL Tree

- Built separate C++ client and server applications on Windows using WinSock for networked file management.
- Server spawns a dedicated thread per client connection and uses an in-memory AVL tree to index millions of files for fast lookup.
- Client supports account login and Linux-style commands to list, create, delete, and modify files on the remote server.

Portfolio: talhazaheer.dev/#projects

EDUCATION

University of Management and Technology (UMT)

Nov 2019 – Aug 2025

Bachelor of Science, Computer Science

Lahore, Pakistan

Professional Freelancing Training Program (NIAIS)

Jun 2024 – May 2025

Diploma, Cyber Security

Lahore, Pakistan

NAVTTC

Apr 2025 – Jul 2025

Certification, Digital Forensic & Cyber Security

Pakistan

ADDITIONAL EXPERIENCE

Skill Bridge Pakistan

Agentic AI Course Instructor

Dec 2025 – May 2026

Lahore, Pakistan

Recorded 24 video lectures on LLMs, prompt engineering, and the Vercel AI SDK (streaming, tool calling, agents); guided students to build and deploy an agentic portfolio chatbot with Next.js on Vercel; taught local LLM development with LM Studio.

Hunarmand Punjab

Course Instructor

Sep 2025 – Oct 2025

Punjab, Pakistan

Delivered structured technical training for a government skills initiative; designed and recorded two courses (Cybersecurity and Penetration Testing, 48 lectures) with quizzes, assignments, and exams.